Experimental verification of multipartite entanglement in the presence of dishonest parties

W. McCutcheon, A.Pappa, B. Bell, A. McMillan, A. Chailloux, T. Lawson, M. Mafu, W. Wadsworth, D. Markham, E. Diamanti, I. Kerenidis, J. Rarity, M. Tame


















Outline

Witnessing Entanglement :

- Trusted and untrusted agents in a network
- Verification protocol for distributed multipartite entanglement

Experimental Implementation :

- Photon pair generation in microstructured photonic crystal fibre
- 3-photon and 4-photon GHZ states

The Model

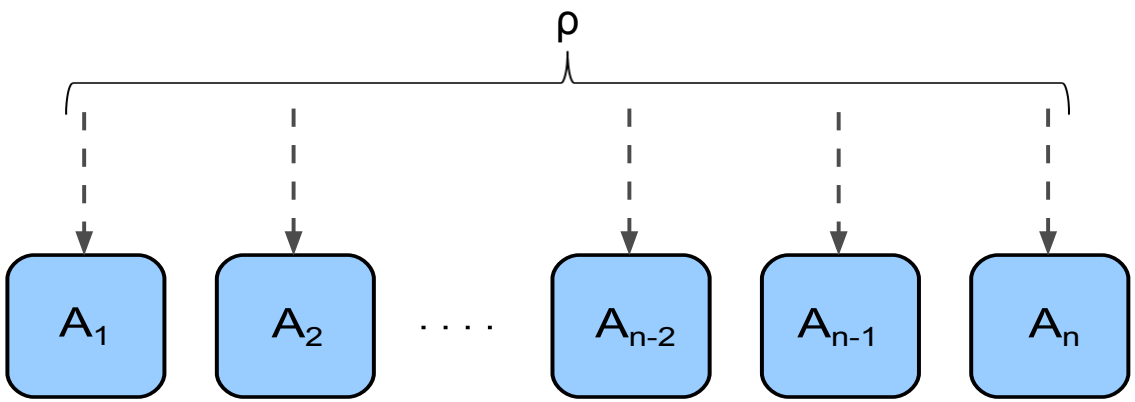
- Untrusted Source: It is supposed to create and share among n parties, the GHZ-state:

$$|G_0^n\rangle = \frac{1}{\sqrt{2}}\left(|0^n\rangle + |1^n\rangle\right)$$

- Parties: Unbounded number of dishonest parties that can collaborate with the source.
- Local Resources: The honest parties have a trusted single-qubit measurement device.
- Network Resources: Pairwise private classical channels.

The Verification Protocol

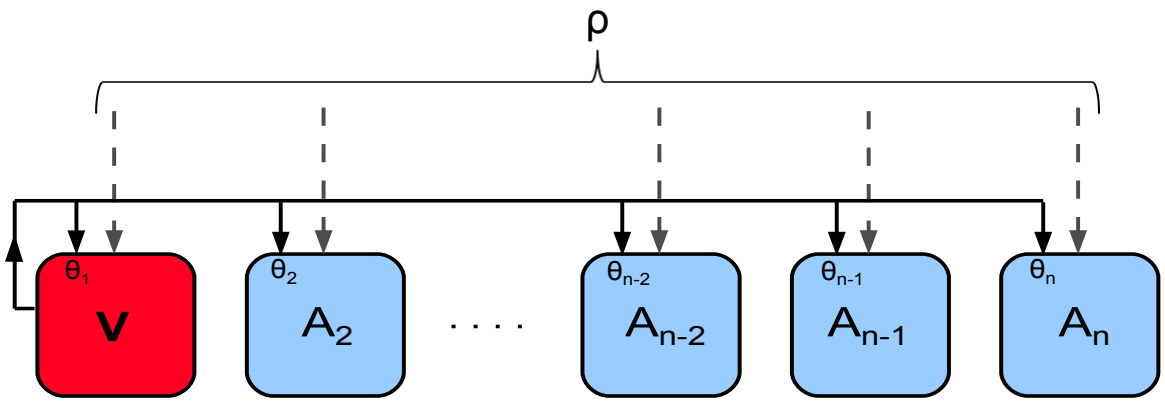
A source shares a state ρ with n parties.



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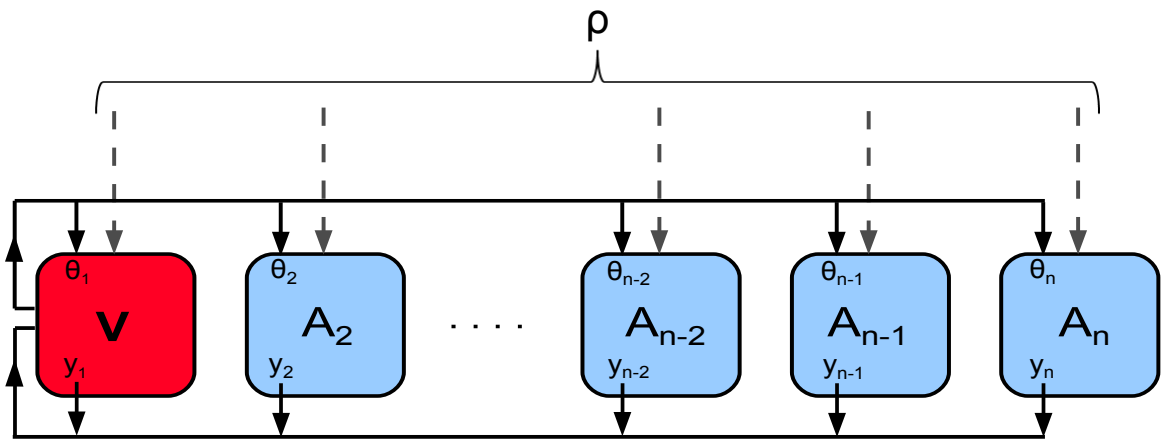
1. The Verifier selects $\theta_j \in_R [0, \pi)$, $\forall j \in [n]$, such that $\sum_{j=1}^{n} \theta_j \equiv 0 \pmod{\pi}$ and sends θ_j to party A_j via a private channel.



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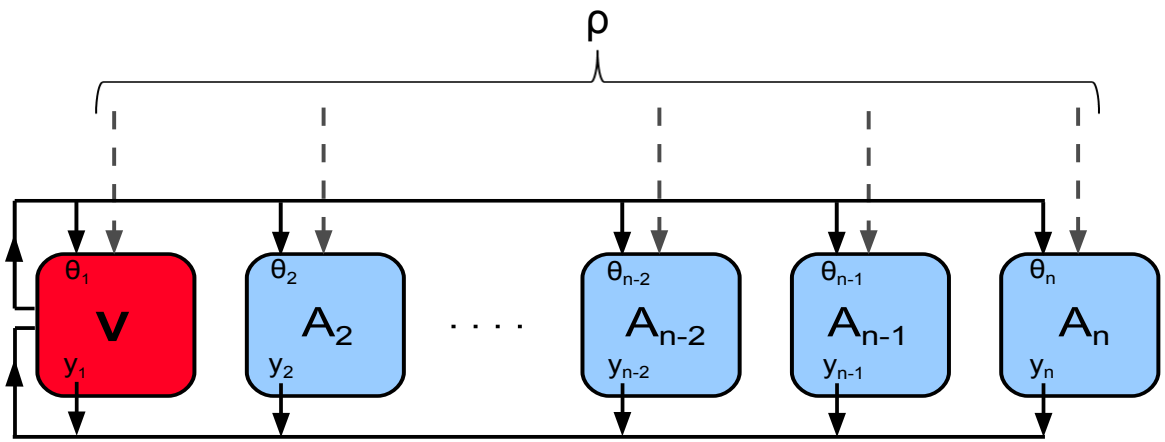
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2. Each party A_j measures in $\{\frac{|0\rangle + e^{i\theta_j}|1\rangle}{\sqrt{2}}, \frac{|0\rangle - e^{i\theta_j}|1\rangle}{\sqrt{2}}\}$, and sends outcome $y_j \in \{0, 1\}$ to the Verifier via the private channel.



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3. The Verifier performs test $T(\rho)$: $\sum_{j=1}^{n} y_j = \frac{1}{\pi} \sum_{j=1}^{n} \theta_j \pmod 2$.



Correctness

The test always succeeds on $|G_0^n\rangle$

The measurement is equivalent to a rotation around the Z axis and a measurement in the Hadamard basis.

From condition $\sum_{j=1}^{n} \theta_j \equiv 0 \pmod{\pi}$, we have two cases:

1. $\sum_{j=1}^{n} \theta_j \equiv 0 \pmod{2\pi} \Rightarrow |G_0^n\rangle$ is "mapped" to $|G_0^n\rangle$.
   It holds that $\sum_{j=1}^{n} y_j \equiv 0 \pmod{2}$.
2. $\sum_{j=1}^{n} \theta_j \equiv \pi \pmod{2\pi} \Rightarrow |G_0^n\rangle$ is "mapped" to $|G_\pi^n\rangle^*$.
   It holds that $\sum_{i=1}^{n} y_i \equiv 1 \pmod{2}$.

$^*|G_\pi^n\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle^n - |1\rangle^n\right)$

Security in the Honest Model

Theorem 1

If $F(\rho) := \langle G_0^n|\rho|G_0^n\rangle$, then $\Pr(T(\rho)) \leq \frac{1}{2}(1 + F(\rho))$.

(Sketch of) Proof:

Test T is equivalent to performing a POVM $\{P_n, I - P_n\}$

$$P_n = |G_0^n\rangle\langle G_0^n| + \frac{1}{2} I_{S_n}$$

I_{S_n}: the projection on the space orthogonal to both $|G_0^n\rangle$ and $|G_\pi^n\rangle$.

$$F(\rho) := \langle G_0^n|\rho|G_0^n\rangle \quad \Rightarrow \quad \rho = F(\rho)|G_0^n\rangle\langle G_0^n| + (1 - F(\rho))\chi,$$

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Security in the Dishonest Model

Theorem 2

If $F'(\rho) := \max_{U_D} F((\mathbb{I}_H \otimes U_D)\rho)$ where U_D is an operator on the space D of the dishonest parties, then $\Pr(T(\rho)) \leq \frac{1}{4}(3 + F'(\rho))$.

(Sketch of) Proof for k honest parties:

For honest angle $\theta = \sum_{j \in H} \theta_j \pmod{\pi}$, a purification exists:

$$|\Psi\rangle = |G_\theta^k\rangle|\Psi_\theta\rangle + |G_{\pi+\theta}^k\rangle|\Psi_{\pi+\theta}\rangle + |\mathcal{X}\rangle$$

The probability to pass the test is:

$$\Pr[T|\Psi\rangle \mid \theta] \leq \frac{1}{2} + \frac{1}{2}\left\| |\Psi_\theta\rangle\langle\Psi_\theta| - |\Psi_{\pi+\theta}\rangle\langle\Psi_{\pi+\theta}| \right\|$$

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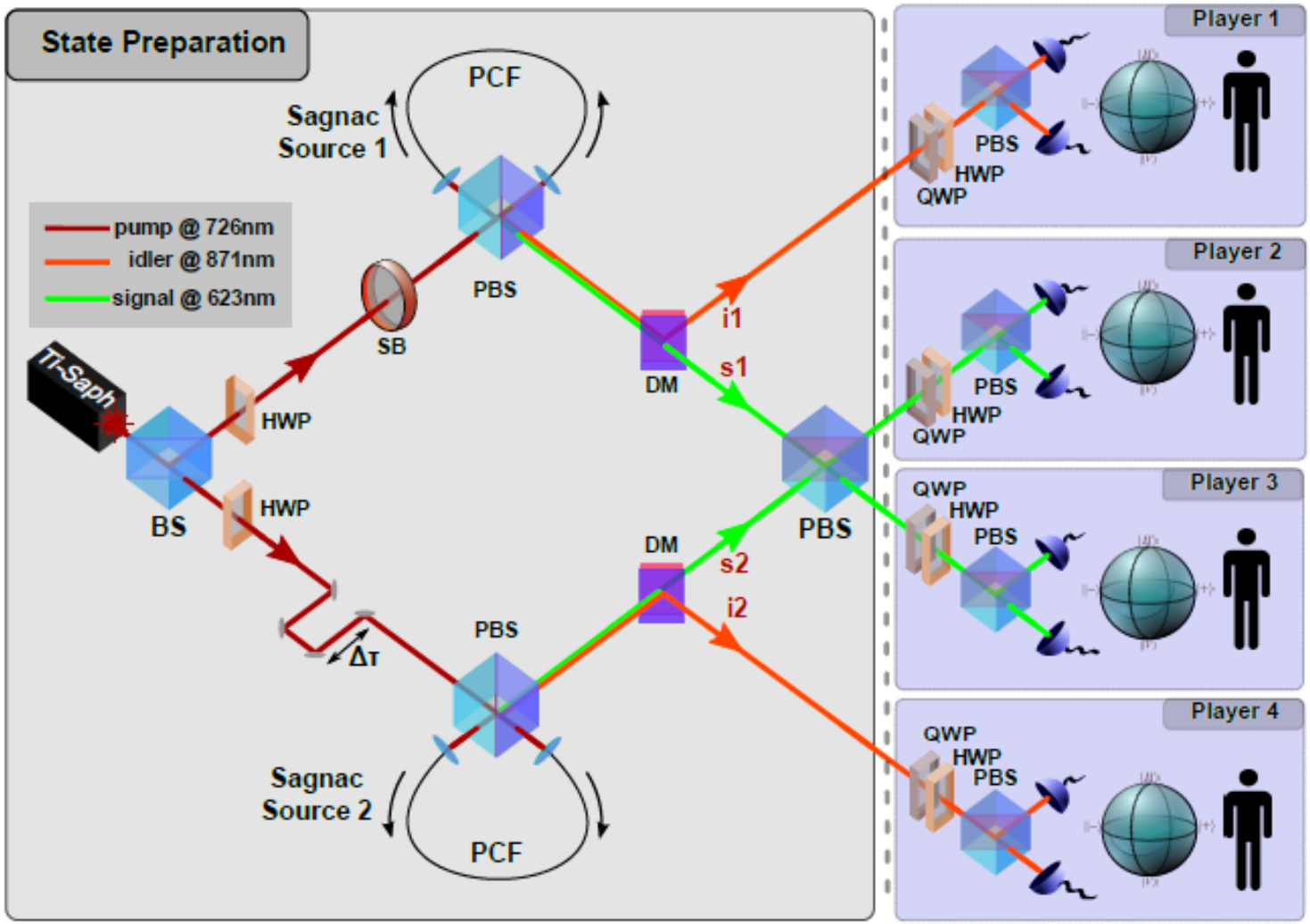
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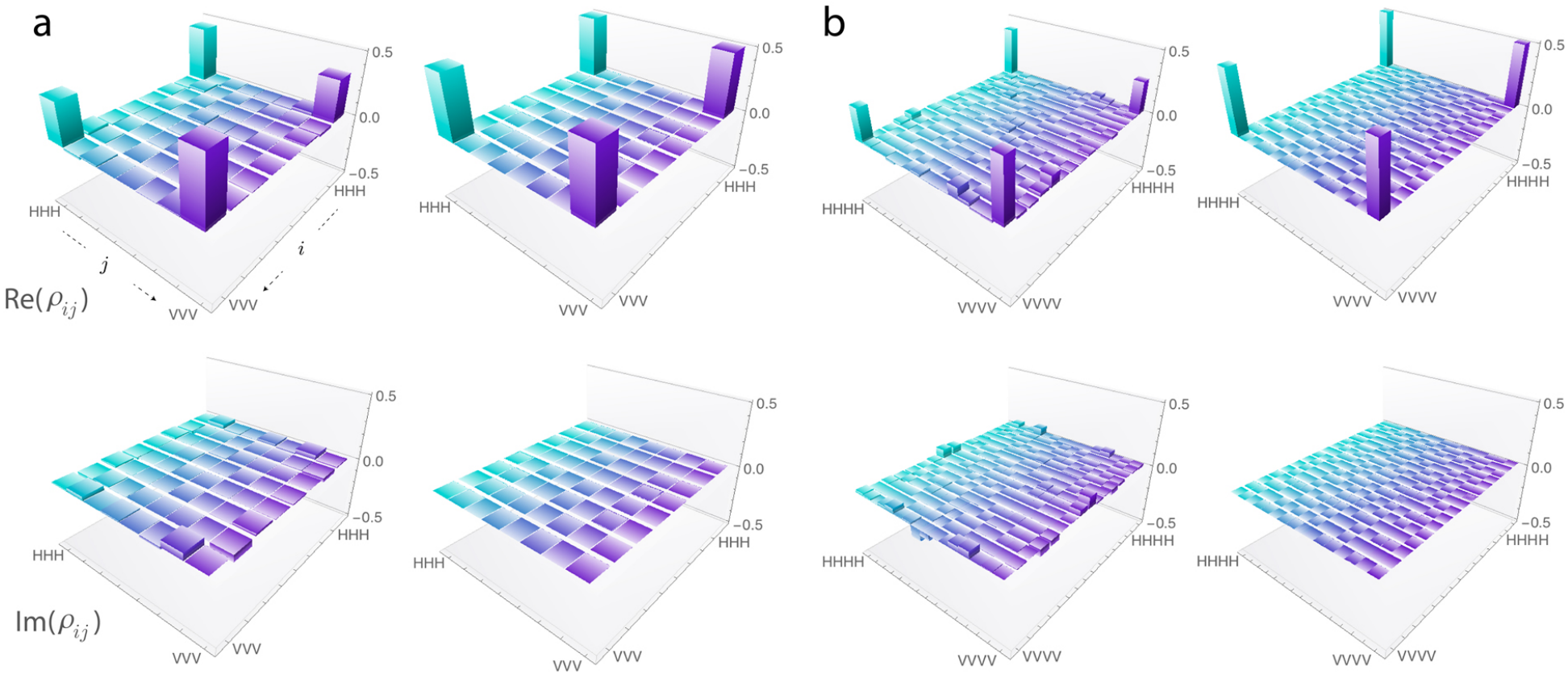
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Implementation



B. Bell, A. Clark, M. S. Tame, M. Halder, J. Fulconis, W. Wadsworth, J. Rarity: *Experimental characterization of photonic fusion using fiber sources*, New J. Phys. 14, 023021 (2012).

Tomography



$F_{GHZ3} = 0.8 \pm 0.01$

$F_{GHZ4} = 0.71 \pm 0.03$

Measurements

GHZ3:

1. XY protocol*
   - Honest Case $\Rightarrow$ entangled state
   - Dishonest Case $\Rightarrow$ separable state
2. θ protocol
   - Honest Case $\Rightarrow$ entangled state
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GHZ4:

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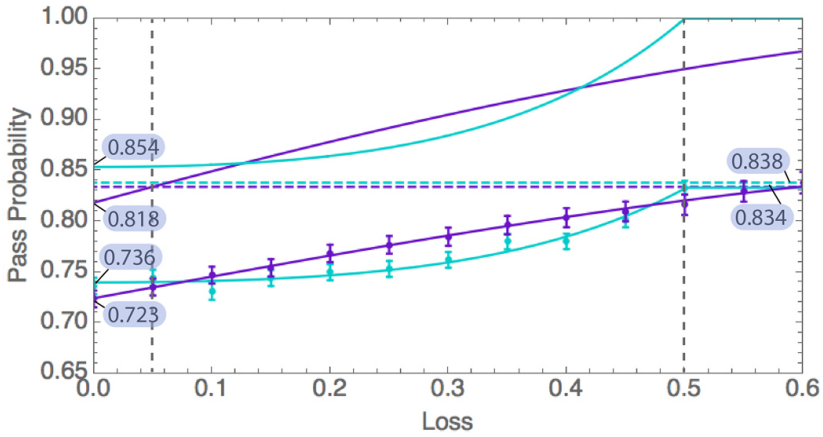
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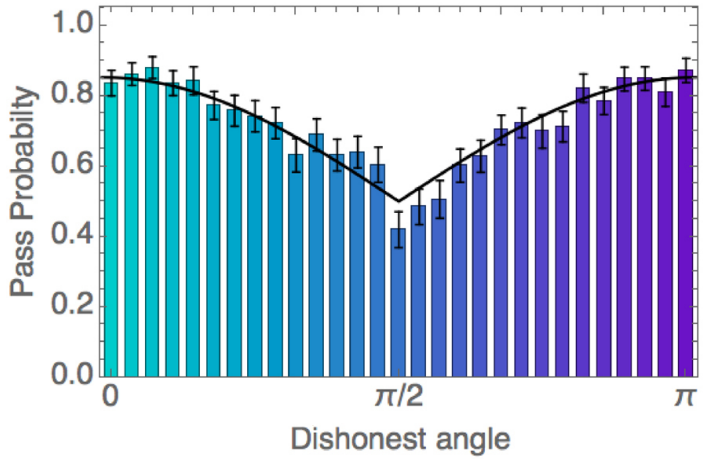
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Results

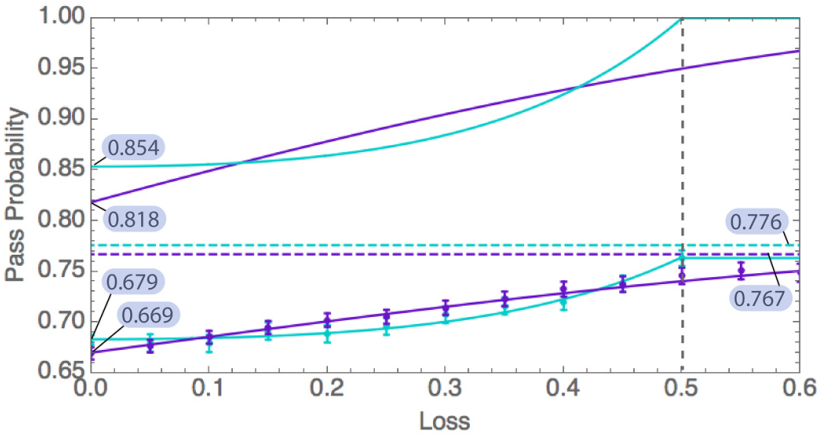
a



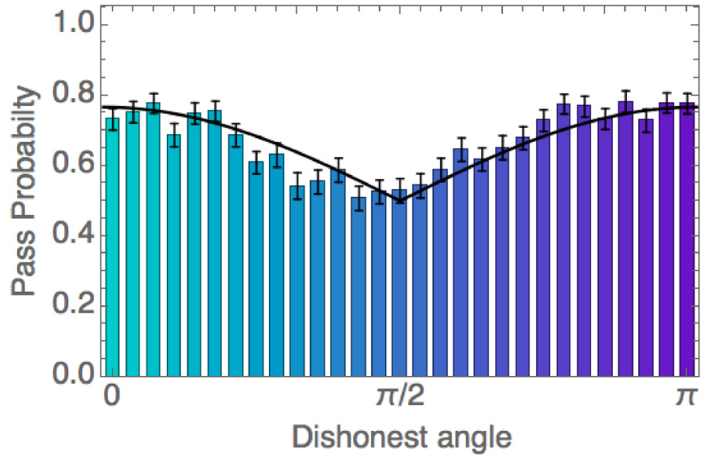
b



c



d



Conclusion / Open Questions

- Protocols for GHZ verification against dishonest parties.
- Fault-tolerance analysis.
- Demonstration with single shot, automated system.
- Extension to other states, such as graph states?
- Use of protocols for Multiparty Computation?